Description of input.ini keywords

1 [imagecata]

1.1 imagefile

- Specifies the main input image file.
- This is typically a FITS file containing the science data to be analyzed.
- Example:

```
imagefile = frame-r-002125-3-0104.fits
```

1.2 whtfile

- Specifies the weight or noise map associated with the input image.
- The interpretation depends on the filename:
  - If it contains rms", it is treated as an RMS map.
  - If it contains weight", it is treated as a weight map.
  - Otherwise, it is assumed to be an RMS map by default.
- Example:

```
whtfile = sigma.fits
```

1.3 sex_cata

- Specifies the catalogue generated by a source detection algorithm.
- Contains detected objects along with their measured properties such as position, flux, and shape parameters.
- Used as the primary detection catalogue for further processing.
- Example:

```
sex\_cata = sex.cat
```

1.4 obj_cata

- Specifies an external catalogue of objects, typically obtained from an online database.
- The expected format includes positional information such as right ascension (RA) and declination (DEC), possibly split into multiple columns.
- This catalogue is used to define target objects for further analysis.
- Example:

  ```
  obj\_cata = input.cat
  ```

  In the case of an image which is not cutout `galcut=False`, then accepted headers are

  ```
  OBJ_ID RA DEC Z STAR
  ```

  ```
  OBJ_ID RA1 RA2 RA3 DEC1 DEC2 DEC3 STAR Z
  ```

  If you are giving PSF file names in the config.ini or giving a file which contains the PSF files, then you can give only

  ```
  OBJ_ID RA DEC Z
  ```

  ```
  OBJ_ID RA1 RA2 RA3 DEC1 DEC2 DEC3 Z
  ```

  If you don't need to convert the structural parameters to physical parameters, you can give the header

  ```
  OBJ_ID RA DEC
  ```

  ```
  OBJ_ID RA1 RA2 RA3 DEC1 DEC2 DEC3
  ```

1.5 rootname

- Defines the base name for all output files generated by the pipeline.
- Output files such as catalogues and images will use this prefix.
- Example:

  ```
  rootname = cl1358
  ```

1.6 `datadir`

- Specifies the directory containing input image files.
- If this keyword is omitted or commented out, the program defaults to using the current working directory.
- Example:

```
datadir = /Users/vinu/github/pymorph_chatgpt/data_large
```

1.7 `outdir`

- Specifies the directory where output data products will be stored.
- If this keyword is omitted or commented out, output files are written to the current working directory.
- Example:

```
outdir = /home/vinu/WorkAll/TNG/morphology/decomposition/
```

2 `[external]`

2.1 `GALFIT_PATH`

- Defines the full path to the GALFIT executable.
- Used by the pipeline to invoke GALFIT for image fitting tasks.
- Example:

```
/usr/local/bin/galfit
```

2.2 `SEX_PATH`

- Specifies the full path to the SExtractor executable.
- Used for source detection and catalogue generation.
- Example:

```
/opt/homebrew/bin/sex
```

3 [psf]

3.1 stargal_prob

- Threshold probability used for star-galaxy classification.
- Objects with classification probability above this value are treated as stars.
- Also used as a selection criterion in fitting modes such as `fit_and_fit`.
- Example:

```
stargal_prob = 0.7
```

3.2 star_size

- Defines the size of the PSF image.
- The PSF cutout size is computed as a multiple of the semi-major axis (SMA) obtained from SExtractor.
- The final PSF size is given by: [PSF size = `star_size` × SMA]
- Example:

```
star_size = 20
```

3.3 psflist

- Specifies a list of PSF images to be used in the analysis.
- Each PSF file name encodes positional information in the format: [`psf_RA1RA2RA3+DEC1DEC2DEC3.fits`]
- The pipeline extracts Right Ascension (RA) and Declination (Dec) from the filename and updates the PSF header accordingly.
- Multiple PSF files are separated by commas.
- Alternatively, if specified as `@filename`, the PSF list will be read from an external file.
- Example:

```
psflist = psf_1447218+0838538.fits, psf_1447219+0828392.fits, ...

psflist = @filename.csv
```

4 [mask]

4.1 manual_mask

- Enables or disables manual masking.
- If set to 1, masking is controlled manually; otherwise automatic masking is used.

4.2 mask_reg

- Defines the radius of the masking region.
- Mask radius is given by: [R_mask = mask_reg × (semi-major axis of neighbour)]

4.3 thresh_area

- Sets the threshold ratio of neighbour area relative to the target object.
- Neighbours with area less than this factor times the object area are masked.

4.4 threshold

- Defines the overlap condition for masking.
- Masking occurs when scaled semi-major axes of object and neighbour overlap.

4.5 no_mask

- If set to 1, masking is completely disabled.

5 [size]

5.1 size_list

- Controls cutout size and resizing behavior.
- Format: [[resize, imsize, fracrad, square, fixsize]]

`resize = 0` and `imsize = 1` when using `galcut = True`. `resize = 1` and `imsize = 0` when using `galcut = False`. Then it will generate size of the image by multipling fracrad with semimajor axis of that object

5.2 `searchrad`

- Search radius in arcseconds for identifying neighbouring objects.

5.3 `nearest_neighbour`

- Boolean flag indicating whether to consider the nearest neighbour during processing. Don't change this value

6 `[cosmology]`

6.1 `pixelscale`

- Pixel scale in arcseconds per pixel.
- Used to convert image measurements into physical units.

7 `[nonparams]`

7.1 `back_extraction_radius`

- Radius used for background estimation.

7.2 `angle`

- Rotation angle (in degrees), typically used for asymmetry calculations.

8 `[modes]`

8.1 `galcut`

- If True, assumes cutout images are already provided.

8.2 `decompose`

- Enables running of the decomposition (e.g., fitting using GALFIT).

8.3 `detail`

- Enables random perturbation of initial fitting parameters.

8.4 `number_random`

- Number of random realizations used when `detail=True`.

8.5 `casgm`

- Enables computation of non-parametric morphology parameters (CASGM).

8.6 `find_and_fit`

- If True, finds and fits all objects within a magnitude range.

9 `[findfit]`

9.1 `maglim`

- Defines faint and bright magnitude limits for object selection.

10 `[mag]`

10.1 `mag_zero`

- Photometric zero point for magnitude calculation.

10.2 `photo_filter`

- Name of the photometric filter used (e.g., SDSS_r).

11 `[galfit]`

11.1 `components`

- Specifies galaxy components to fit (e.g., bulge, disk, bar).

11.2 `devauc`

- If True, uses de Vaucouleurs profile ($n = 4$) for bulge.

11.3 `fitting`

- Flags indicating which components or parameters are free during fitting. The first for fitting bulge centre, second for disk center, next is for sky, next is for centre of the bar

11.4 bbox, barbox

- Initial boxiness parameters for bulge and bar components. If bbox=-99 it will not fit bulge boxiness. Also, barbox=-99 will not find bar boxiness

12 [version]

12.1 galfitv

- Version of GALFIT used.

12.2 pymorphv

- Version of the pipeline.

13 [diagnosis]

13.1 chi2nu_limit

- Upper limit of reduced chi-square for a good fit.

13.2 goodness_limit

- Sigma threshold for statistical goodness of fit.

13.3 center_deviation_limit

- Maximum allowed deviation between initial and fitted center.

13.4 do_plot

- Enables plotting of results.

14 [params_limit]

14.1 NMag

- Magnitude variation range relative to initial estimate.

14.2 `NRadius`

- Maximum radius scaling factor relative to initial estimate.

14.3 `LN`, `UN`

- Lower and upper limits of S'ersic index.

14.4 `center_constrain`

- Maximum allowed positional shift during fitting.

15 `[sextractor]`

- Used to determine the initial configuration parameters. Certain parameters, such as GAIN, must be adjusted for each telescope

Non-parametric parameters

Non-parametric morphology measures are widely used to quantify the structure of galaxies without assuming specific models. The most commonly used parameters are Concentration (C), Asymmetry (A), Clumpiness (S), Gini coefficient (G), and M_{20}.

1 Concentration (C)

The concentration index measures how centrally concentrated the light is:

$$C = 5\log\left(\frac{r_{80}}{r_{20}}\right) \tag{1}$$

where r_{20} and r_{80} are the radii containing 20

2 Asymmetry (A)

Asymmetry quantifies how symmetric a galaxy is under 180° rotation:

$$A = \frac{\sum |I(x,y) - I_{180}(x,y)|}{\sum |I(x,y)|} - A_{\mathrm{bkg}} \tag{2}$$

where I_{180} is the image rotated by 180°, and A_{bkg} accounts for background asymmetry.

3 Clumpiness (S)

Clumpiness measures the fraction of light in high-frequency structures:

$$S = \frac{\sum |I(x,y) - I_S(x,y)|}{\sum |I(x,y)|} - S_{\mathrm{bkg}} \tag{3}$$

where I_S is the smoothed image.

4 Gini Coefficient (G)

The Gini coefficient measures the inequality in the distribution of pixel fluxes:

$$G = \frac{1}{\bar{X}n(n-1)}\sum_{i=1}^{n}(2i-n-1)X_i \tag{4}$$

where X_i are the pixel fluxes sorted in increasing order, $\bar{X}$ is the mean flux, and n is the number of pixels.

5 M_{20}

M_{20} is the normalized second-order moment of the brightest 20

$$M_{20} = \log_{10}\left(\frac{\sum_i M_i}{M_{\text{tot}}}\right) \tag{5}$$

where the sum is over the brightest pixels containing 20

$$M_i = f_i\left[(x_i - x_c)^2 + (y_i - y_c)^2\right] \tag{6}$$

$$M_{\text{tot}} = \sum_i M_i \tag{7}$$

Here f_i is the flux in pixel i, and (x_c, y_c) is the galaxy center.

6 Conclusion

These parameters together provide a powerful, model-independent description of galaxy morphology and are widely used in galaxy evolution studies.